4 The cubic equation $x^3+bx^2+cx-1=0$, where $b$ and $c$ are constants, has roots $\alpha$, $\beta$, $\gamma$.

It is given that the matrix $\begin{pmatrix} 1 & \alpha & \beta \\ \alpha & 1 & \gamma \\ \beta & \gamma & 1 \end{pmatrix}$ is singular.

**(a)** Show that $\alpha^2+\beta^2+\gamma^2=3$. [4]

**(b)** It is given that $\alpha^3 + \beta^3 + \gamma^3 = 3$ and that the constants $b$ and $c$ are positive.

Find the values of $b$ and $c$. [6]